FP3: Greg

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Original Goal

- General purpose processor
- Implement pipeline instead of single cycle
- Use variable length instructions
- 16, 16-bit registers
- 16-bit memory

Implementation

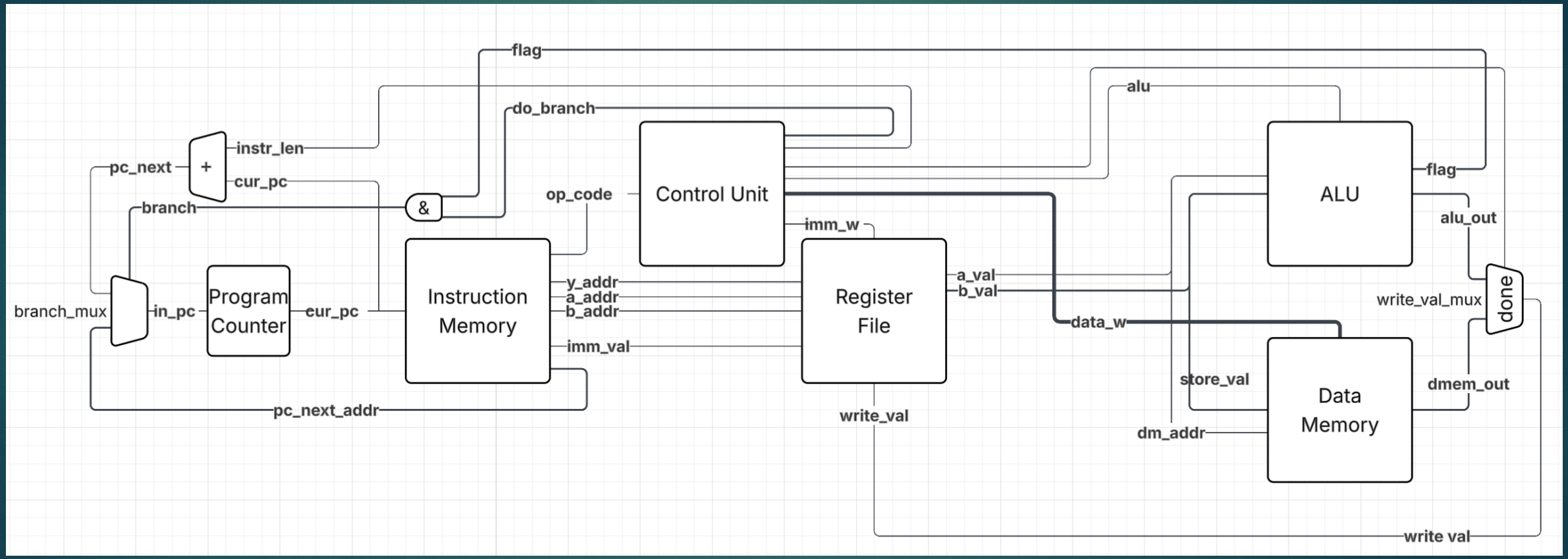
- Greg
- General purpose processor
- Single cycle processor
- Used variable length instructions
- Assembler written in .lua
- 16, 16-bit registers
- 16-bit memory

Assembly Language Instructions

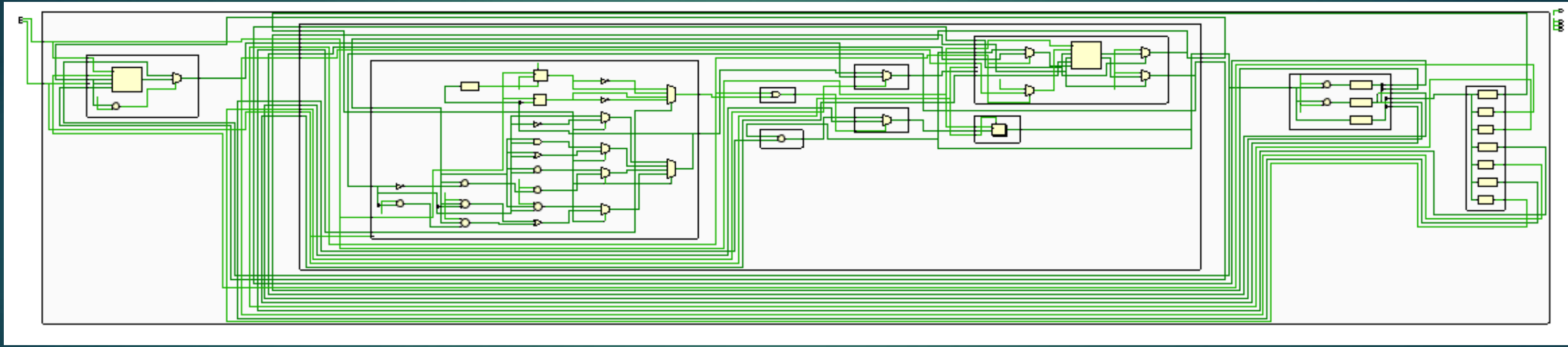
Control Signals

Mnemonic	Opcode	ALU	DATA_W	FLAG_SEL	IMM_W	REG_W_SRC	DO_BRANCH	INSTR_LEN
nop	0000	-	0	-	-	-	-	01
and	0001	001	0	-	0	0	0	10
add	0010	010	0	-	0	0	0	10
lrs	0011	011	0	-	0	0	0	10
not	0100	100	0	-	0	0	0	10
or	0101	101	0	-	0	0	0	10
sub	0110	110	0	-	0	0	0	10
rot	0111	111	0	-	0	0	0	10
j	1000	-	0	-	0	1	0	11
l	1010	-	0	-	0	1	0	10
imm	1001	-	0	-	1	0	0	11
s	1011	-	1	-	0	0	0	10
cbz	1100	-	0	00	0	1	1	11
cbnz	1101	-	0	01	0	1	1	11
cbneg	1110	-	0	10	0	1	1	11
cbnneg	1111	-	0	11	0	1	1	11

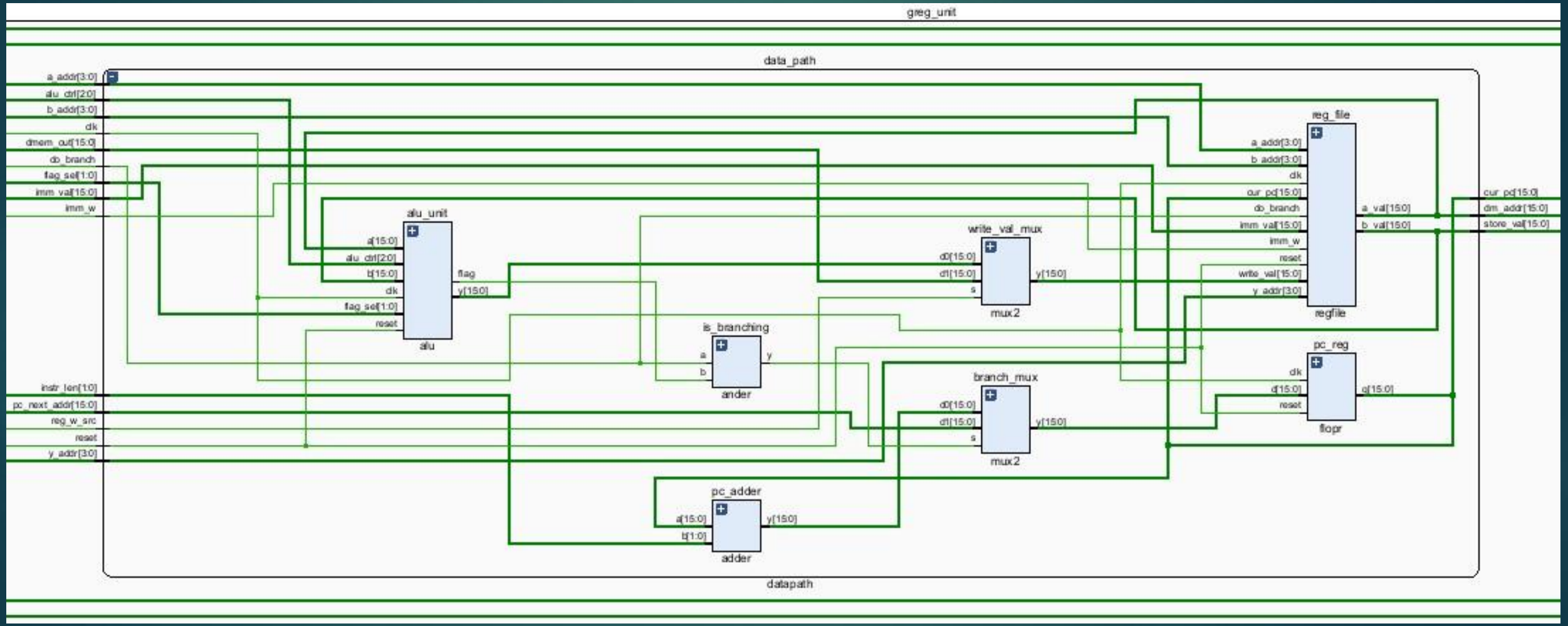
Greg Diagram



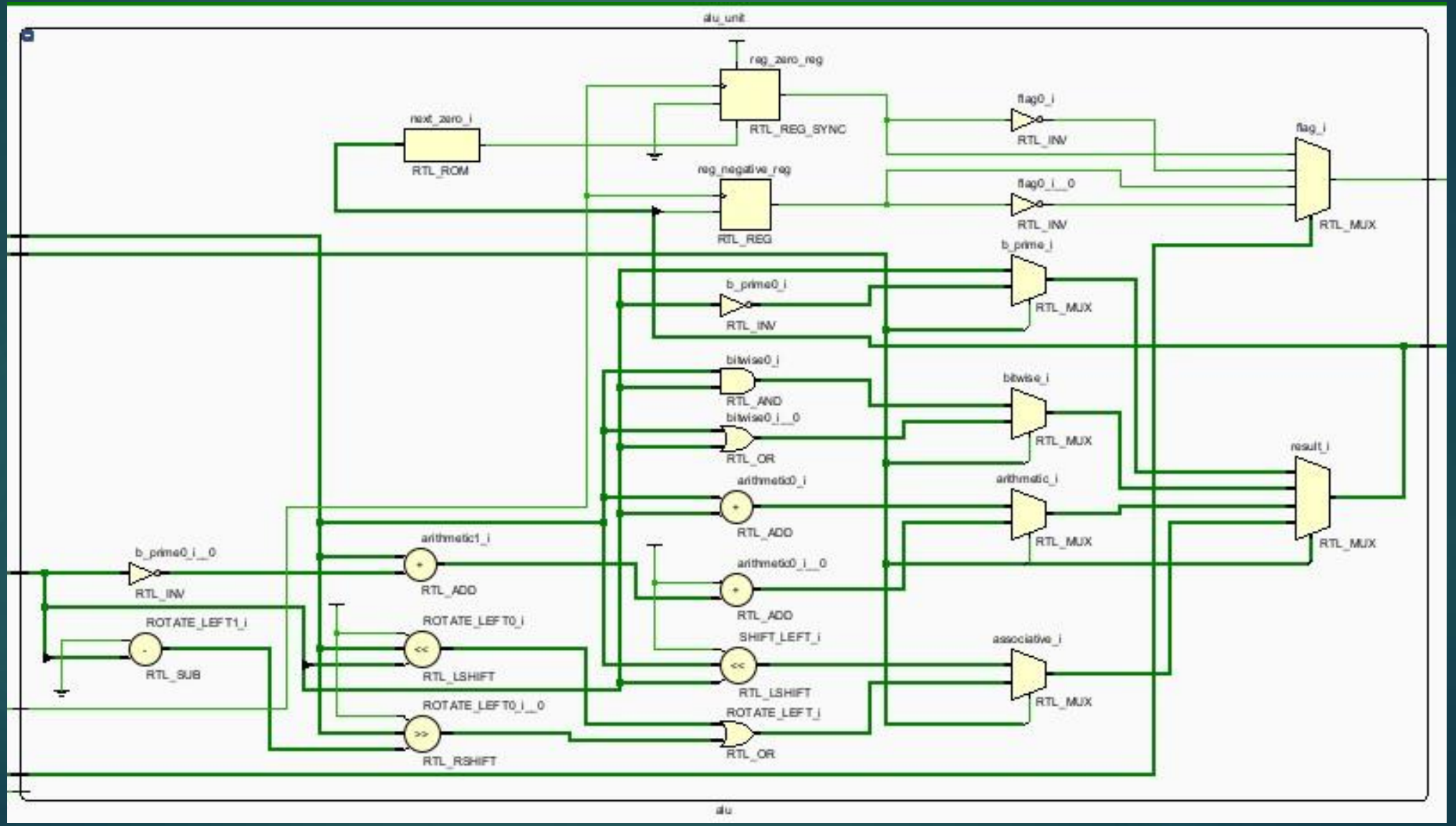
Greg Design – Full Scale



Greg Design – Datapath



Greg Design – ALU



I/O Design

- Attempted implementing a basic I/O using LEDS, BTN and SW on FPGA
- Ran into several bugs
- Planned on implementing years alive test program

Project Reflection

- Challenges:
  - Understanding .vhdl
  - Conceptualizing what was required in .vhdl was a large roadblock
  - Implementing variable length instructions
- Lessons:
  - Steady progress is great and helps the flow of the project
  - Utilize resources (Whitgit examples, Kent office hours, etc.)
  - Vivado structure (internal signaling)
  - More efficient in pairs (value of different coding styles & views)

Questions?